

Lab 7 Report

Proportional–Velocity (PV) Position Control of the Quanser QUBE-Servo

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Abstract

This lab focuses on using a proportional–velocity (PV) controller for QUBE-Servo position control. A Simulink model was developed and tested to evaluate the effects of proportional and derivative gains on system performance. Controller gains were designed using second-order system specifications and implemented on the servo. The resulting position response was analyzed to verify that the desired peak time, overshoot, and overall stability were achieved.

1 System Built

The system built in this lab was a proportional–velocity (PV) position control system for the Quanser QUBE-Servo. The system was implemented in Simulink and consisted of a reference signal generator, a proportional controller, a velocity feedback path with a high-pass filter, and the QUBE-Servo plant. A square-wave position command was used as the reference input, while the measured servo position was fed back to the controller. The proportional gain was used to improve tracking performance, while the derivative gain provided damping and reduced overshoot. The completed system was tested to evaluate position-control performance and to verify the controller gains designed from second-order system specifications.

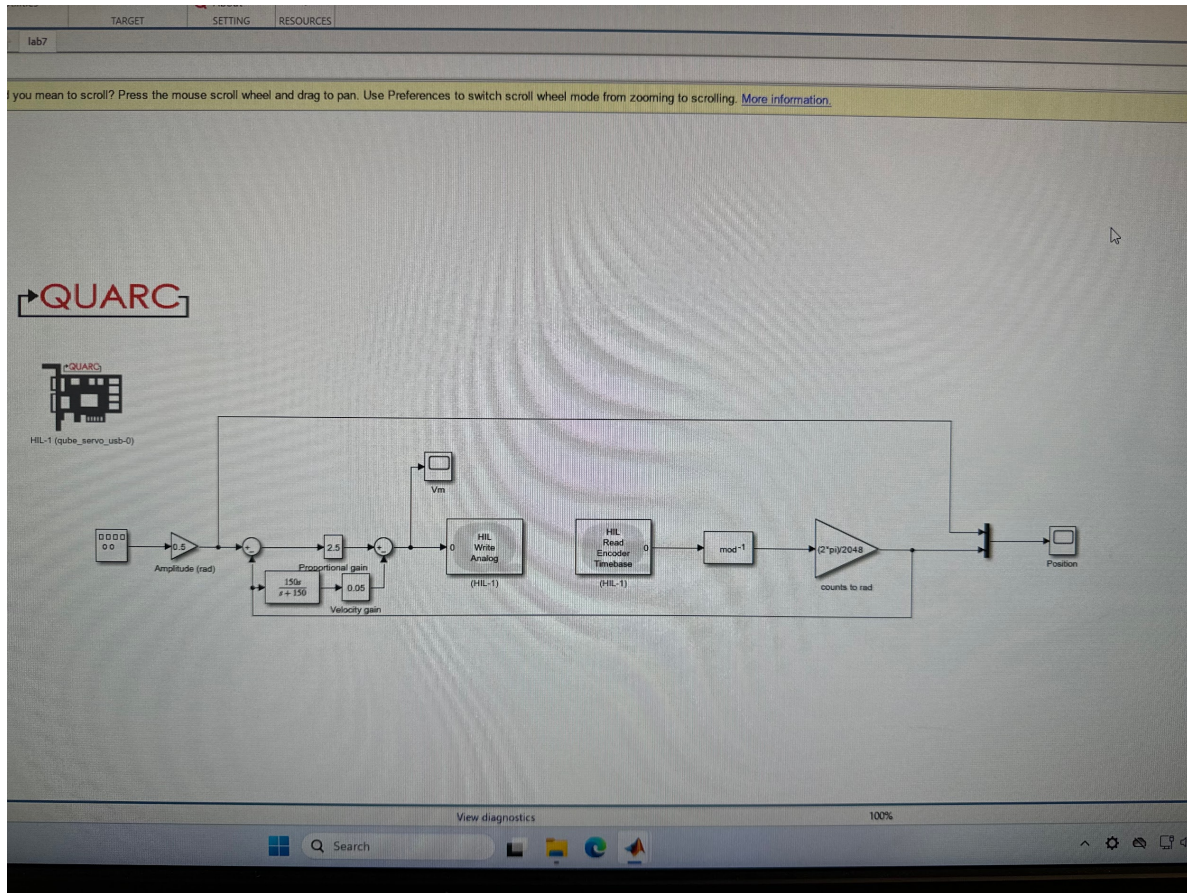


Figure 1: QUARC/Simulink PV position control model for the QUBE-Servo. The proportional gain K_p acts on the position error while the velocity feedback path uses a high-pass filter ($15s/(s + 150)$) and derivative gain K_d for damping; the controller output drives the motor through the HIL Write Analog block.

2 In-Lab Exercises and Results

2.1 Effect of the Proportional and Derivative Gains

The motor voltage command and the position-tracking response were observed while sweeping the proportional gain K_p and the derivative gain K_d . Figure 2 shows the motor voltage for the square-wave command, and Figure 3 shows the reference and measured position together.

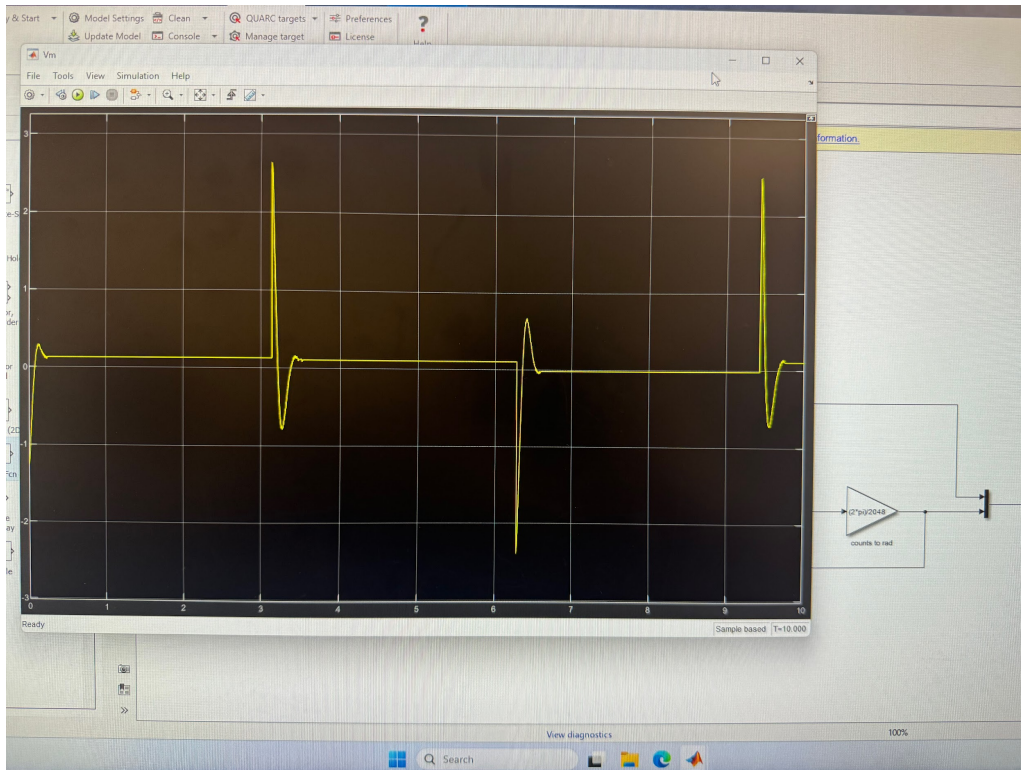


Figure 2: Motor voltage command V_m for the square-wave position reference.

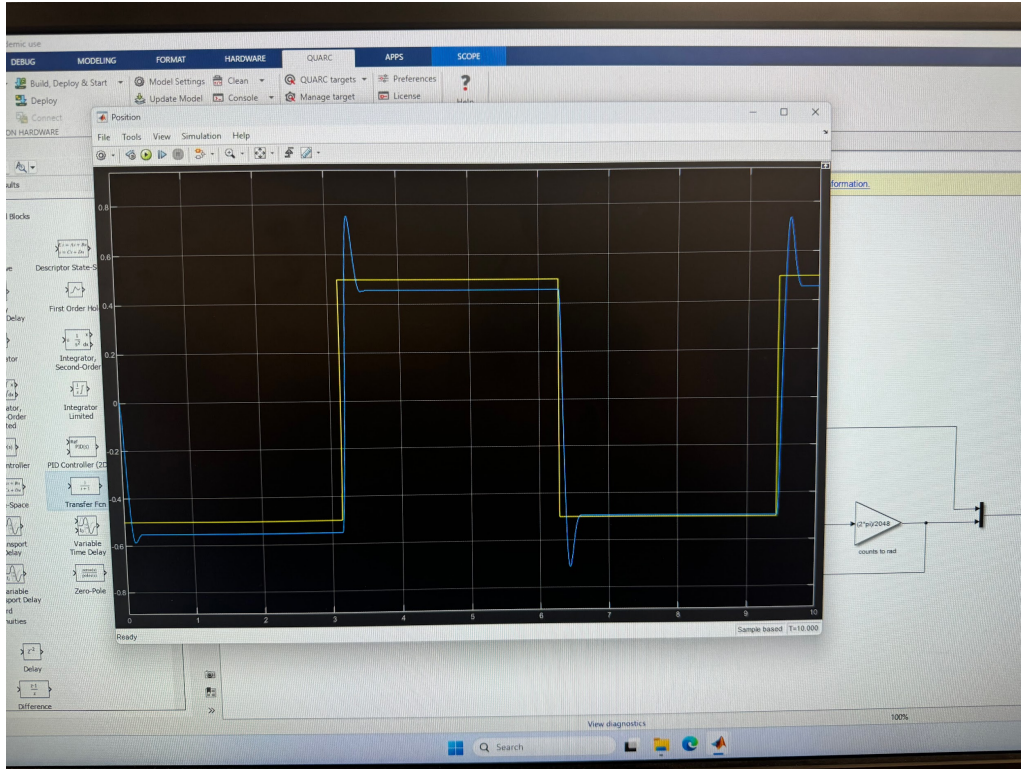


Figure 3: Position-tracking response: the square-wave reference (yellow) and the measured servo position (blue) following it.

Sweeping the proportional gain produced the following observations:

- $K_p = 1$: the amplitude is not as high — the previous response was around 2.5, while this is around 1.
- $K_p = 2$: the output has more oscillations; at least, the smaller oscillations are now larger.
- $K_p = 3$: even more oscillations than when $K_p = 2$.
- $K_p = 4$: even more oscillations than when $K_p = 3$.

As K_p increases there are more oscillations, which shows that the servo is moving more frequently the higher the value of K_p . In contrast, as K_d increases, the output result becomes more similar to the reference.

2.2 Controller Gain Design

Matching the QUBE-Servo closed-loop transfer function to the standard second-order form yields the proportional and derivative gains

$$K_p = \frac{\tau \omega_n^2}{K}, \quad K_d = \frac{2\zeta \omega_n \tau - 1}{K}. \quad (1)$$

Using the model parameters $\omega_n = 32.3$, $\zeta = 0.76$, $K = 23.2$, and $\tau = 0.13$,

$$K_p = \frac{(0.13)(32.3)^2}{23.2} = 5.85, \quad (2)$$

$$K_d = \frac{2(0.76)(32.3)(0.13) - 1}{23.2} = 0.232. \quad (3)$$

Therefore the designed gains are

$$K_p = 5.85 \text{ V/rad} \quad \text{and} \quad K_d = 0.232 \text{ V/(rad/s)}. \quad (4)$$

3 Analysis

Results analysis. The PV controller successfully controlled the position of the QUBE-Servo and demonstrated the effects of proportional and derivative feedback on system performance. Increasing the proportional gain improved the speed of the response and reduced tracking error, but excessive proportional gain increased overshoot and oscillatory behavior. Increasing the derivative gain added damping to the system, reducing overshoot and improving stability. Using the calculated gains, the servo response closely followed the reference position with a fast rise time and minimal overshoot. Overall, the experimental results were consistent with the theoretical second-order system model and demonstrated that PV control can effectively improve transient response while maintaining stable operation.

Issues encountered. None encountered.

4 Conclusion

In this lab, a proportional–velocity (PV) controller was designed and implemented to control the position of the QUBE-Servo system. The effects of proportional and derivative gains on the transient response were investigated, and controller gains were calculated using second-order system design specifications. Experimental testing showed that proportional gain increased response speed while derivative gain improved damping and reduced overshoot. The final controller achieved stable position tracking and demonstrated the effectiveness of PV control for servo position regulation. Overall, the lab provided practical experience in control system design, tuning, and performance analysis using a real electromechanical system.